

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**

Product Identifier

|                      |                      |
|----------------------|----------------------|
| Perihalan Produk:    | <u>Platinum foil</u> |
| Product Description: | <u>Platinum foil</u> |
| Cat No. :            | 42456                |
| CAS No               | 7440-06-4            |
| Molecular Formula    | Pt                   |

Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

|         |                                                                                                                                                                                                                                                       |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Company | Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd<br>Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,<br>No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,<br>Selangor Darul Ehsan, Malaysia.<br>Main line: +60 3-5525 7888 |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Supplier

|                |                             |
|----------------|-----------------------------|
| E-mail address | Enquiry.my@thermofisher.com |
|----------------|-----------------------------|

|                            |                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Emergency Telephone Number | Tel: +03-5525 7888<br>CHEMTREC Malaysia <b>1-800-815-308</b> (Malay)<br>CHEMTREC Malaysia (Kuala Lumpur) <b>+(60)-327884561</b> (Malay) |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|

**SECTION 2: HAZARDS IDENTIFICATION**

Classification of the substance or mixture

Label Elements

**Hazard Statements**

Other Hazards

This product does not contain any known or suspected endocrine disruptors

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

# SAFETY DATA SHEET

Platinum foil

Revision Date 27-Mar-2025

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Platinum  | 7440-06-4 | <= 100   |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

**Eye Contact** Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact** Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Ingestion** Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Inhalation** Remove to fresh air. Get medical attention immediately if symptoms occur.

**Self-Protection of the First Aider** No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

**Suitable Extinguishing Media**  
approved class D extinguishers.

**Extinguishing media which must not be used for safety reasons**  
Water may be ineffective.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Platinum oxide.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. No special precautions required.

# SAFETY DATA SHEET

Platinum foil

Revision Date 27-Mar-2025

## Environmental precautions

Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

## Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Pick up and transfer to properly labelled containers.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry place. Keep away from acids.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV                | OSHA PEL                           |
|-----------|----------|--------------------------|------------------------------------|
| Platinum  |          | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup> |

| Component | European Union | The United Kingdom                                                 | Germany                                     |
|-----------|----------------|--------------------------------------------------------------------|---------------------------------------------|
| Platinum  |                | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr | TWA: 1 mg/m <sup>3</sup> (8 Stunden). AGW - |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### **Eye Protection**

Wear safety glasses with side shields (or goggles)

#### **Hand Protection**

No special protective equipment required

#### **Skin and body protection**

Long sleeved clothing

#### **Respiratory Protection**

No protective equipment is needed under normal use conditions

#### **Recommended Filter type:**

Particle filter

# SAFETY DATA SHEET

Platinum foil

Revision Date 27-Mar-2025

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                         |                                     |                                          |
|-----------------------------------------|-------------------------------------|------------------------------------------|
| Appearance                              | Silver                              |                                          |
| Physical State                          | Solid                               |                                          |
| Odor                                    | Odorless                            |                                          |
| Odor Threshold                          | No data available                   |                                          |
| pH                                      | Not applicable                      |                                          |
| Melting Point/Range                     | 1769 - 1773 °C / 3216.2 - 3223.4 °F |                                          |
| Softening Point                         | No data available                   |                                          |
| Boiling Point/Range                     | 3825 °C / 6917 °F                   | @ 7600 mmHg                              |
| Flash Point                             | No information available            | <b>Method -</b> No information available |
| Evaporation Rate                        | Not applicable                      | Solid                                    |
| Flammability (solid,gas)                | No information available            |                                          |
| Explosion Limits                        | No data available                   |                                          |
| Vapor Pressure                          | No data available                   |                                          |
| Vapor Density                           | Not applicable                      | Solid                                    |
| Specific Gravity / Density              | 21.4500                             |                                          |
| Bulk Density                            | No data available                   |                                          |
| Water Solubility                        | Insoluble in water                  |                                          |
| Solubility in other solvents            | No information available            |                                          |
| Partition Coefficient (n-octanol/water) |                                     |                                          |
| Autoignition Temperature                | No data available                   |                                          |
| Decomposition Temperature               | No data available                   |                                          |
| Viscosity                               | Not applicable                      | Solid                                    |
| Explosive Properties                    | No information available            |                                          |
| Oxidizing Properties                    | No information available            |                                          |
| Molecular Formula                       | Pt                                  |                                          |
| Molecular Weight                        | 195.08                              |                                          |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

# SAFETY DATA SHEET

Platinum foil

Revision Date 27-Mar-2025

Stable under normal conditions.

## Possibility of Hazardous Reactions

|                                 |                               |
|---------------------------------|-------------------------------|
| <b>Hazardous Polymerization</b> | No information available.     |
| <b>Hazardous Reactions</b>      | None under normal processing. |

## Conditions to Avoid

None known.

## Incompatible Materials

Acids.

## Hazardous Decomposition Products

Platinum oxide.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

|                                        |                                                           |
|----------------------------------------|-----------------------------------------------------------|
| (a) acute toxicity;                    |                                                           |
| Oral                                   | No data available                                         |
| Dermal                                 | No data available                                         |
| Inhalation                             | No data available                                         |
| (b) skin corrosion/irritation;         | No data available                                         |
| (c) serious eye damage/irritation;     | No data available                                         |
| (d) respiratory or skin sensitization; |                                                           |
| Respiratory                            | No data available                                         |
| Skin                                   | No data available                                         |
| (e) germ cell mutagenicity;            | No data available                                         |
| (f) carcinogenicity;                   | No data available                                         |
|                                        | There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;             | No data available                                         |
| (h) STOT-single exposure;              | No data available                                         |
| (i) STOT-repeated exposure;            | No data available                                         |

# SAFETY DATA SHEET

Platinum foil

Revision Date 27-Mar-2025

**Target Organs** No information available.

**(j) aspiration hazard;** Not applicable  
Solid

**Symptoms / effects, both acute and delayed** No information available.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

**Persistence and degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
**Persistence** Insoluble in water, May persist.  
**Degradability** Not relevant for inorganic substances.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential** May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

**Mobility in soil** Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Dispose of in accordance with local regulations

**Contaminated Packaging** Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

**Road and Rail Transport** Not regulated

# SAFETY DATA SHEET

Platinum foil

Revision Date 27-Mar-2025

**IATA** Not regulated

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories** X = listed

| Component | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------|-----------|------|-----|-------|------|------|-------|------|----------|
| Platinum  | 231-116-1 | X    | X   | X     | X    |      | X     | X    | KE-28808 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

27-Mar-2025

Not applicable.

# SAFETY DATA SHEET

Platinum foil

Revision Date 27-Mar-2025

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**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## **Disclaimer**

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**End of Safety Data Sheet**